

APPENDIX B

Quick Introduction to R

This is just a brief introduction to R. See the preface for recommendations about how to learn more about R.

B.1 Reading the Data In

The first step is to read the data in. You can use the `read.table()` or `scan()` functions to read data in from outside R. You can also use the `data()` function to access data already available within R:

```
> data (stackloss)
> stackloss
  Air.Flow Water.Temp Acid.Conc. stack.loss
1      80         27         89         42
2      80         27         88         37
...stuff deleted...
21     70         20         91         15
```

Type:

```
> help (stackloss)
```

to see more information about the data. We can check the dimension of the data:

```
> dim (stackloss)
[1] 21 4
```

There are 21 rows and four columns.

B.2 Numerical Summaries

One easy way to get the basic numerical summaries is:

```
> summary (stackloss)
  Air.Flow  Water.Temp  Acid.Conc.  stack.loss
Min.   :50.0 Min.   :17.0 Min.   :72.0 Min.    : 7.0
1st Qu.:56.0 1st Qu.:18.0 1st Qu.:82.0 1st Qu.:11.0
Median :58.0 Median :20.0 Median :87.0 Median :15.0
Mean   :60.4 Mean   :21.1 Mean   :86.3 Mean   :17.5
3rd Qu.:62.0 3rd Qu.:24.0 3rd Qu.:89.0 3rd Qu.:19.0
Max.   :80.0 Max.   :27.0 Max.   :93.0 Max.   :42.0
```

We can also compute these numbers separately:

```
> stackloss$Air.Flow
[1] 80 80 75 62 62 62 62 62 58 58 58 58 58 58 50 50 50 50 50 56
[21] 70
> mean (stackloss$Ai)
[1] 60.429
> median(stackloss$Ai)
[1] 58
> range (stackloss$Ai)
[1] 50 80
> quantile (stackloss$Ai)
 0%  25%  50%  75% 100%
 50   56   58   62   80
```

We can get the variance and standard deviation:

```
> var (stackloss$Ai)
[1] 84.057
> sd (stackloss$Ai)
[1] 9.1683
```

We might also want the correlations:

```
> cor (stackloss)
           Air.Flow Water.Temp Acid.Conc. stack.loss
Air.Flow    1.00000    0.78185    0.50014    0.91966
Water.Temp  0.78185    1.00000    0.39094    0.87550
Acid.Conc.  0.50014    0.39094    1.00000    0.39983
stack.loss  0.91966    0.87550    0.39983    1.00000
```

B.3 Graphical Summaries

We can make histograms and boxplots and specify the labels:

```
> hist (stackloss$Ai)
> hist (stackloss$Ai, main="Histogram of Air Flow",
      xlab="Flow of cooling air")
> boxplot (stackloss$Ai)
```

Scatterplots are also easily constructed:

```
> plot (stackloss$Ai, stackloss$W)
> plot (Water.Temp ~ Air.Flow, stackloss, xlab="Air Flow"
      ylab="Water Temperature")
```

We can make a scatterplot matrix:

```
> plot (stackloss)
```

We can put several plots in one display:

```
> par(mfrow=c (2, 2))
> boxplot (stackloss$Ai)
> boxplot (stackloss$Wa)
> boxplot (stackloss$Ac)
> boxplot (stackloss$S)
> par(mfrow=c(1, 1))
```

where the final command causes a return to one plot per display.

B.4 Selecting Subsets of the Data

The second row:

```
> stackloss [2,]

  Air.Flow Water.Temp Acid.Conc. stack.loss
2         80         27         88         37
```

The third column:

```
> stackloss [,3]
[1] 89 88 90 87 87 87 93 93 87 80 89 88 82 93 89 86 72 79 80 82
[21] 91
```

The (2, 3) element:

```
> stackloss [2, 3]
[1] 88
```

`c()` is a function for making vectors—for example:

```
> c (1, 2, 4)
[1] 124
```

Select the first, second and fourth rows:

```
> stackloss [0 (1, 2, 4),]
  Air.Flow Water.Temp Acid.Conc. stack.loss
1         80         27         89         42
2         80         27         88         37
4         62         24         87         28
```

The: operator is good for making sequences—for example:

```
> 3:11
[1] 3 4 5 6 7 8 9 10 11
```

We can select the third through sixth rows:

```
> stackloss [3:6,]
  Air.Flow Water.Temp Acid.Conc. stack.loss
3      75         25        90         37
4      62         24        87         28
5      62         22        87         18
6      62         23        87         18
```

We can use “-” to indicate “everything but”—that is all the data except the first two columns:

```
> stackloss[,-c (1, 2)]
  Acid.Conc. stack.loss
1          89         42
2          88         37
...
21         91         15
```

We may also want to select the subsets on the basis of some criterion—for example, those cases which have an air flow greater than 72:

```
> stackloss[stackloss$Ai > 72,]
  Air.Flow Water.Temp Acid.Conc. stack.loss
1      80         27        89         42
2      80         27        88         37
3      75         25        90         37
```

B.5 Learning More about R

While running R you can get help about a particular command, for example, if you want help about the `boxplot ( )` command, just type `help (boxplot)`. If you do not know what the name of the command is that you want to use, then type:

```
help.start ( )
```

and then browse. You will be able to pick up the language from the examples in the text and from the help pages.